sum, min, max, and avg to perform calculations on data. Bucket aggregations help us categorize data into buckets or ranges. Finally, pipeline aggregations permit us to chain aggregations: that is take metric or bucket aggregations and create new aggregations.

Elasticsearch provides a lot of aggregations out of the box. We get familiar with each type before jumping into a few to get our hands dirty. Note that it is impractical to look at them individually due to space restrictions, but learning the concepts and applying them to several common aggregations is more important than documenting every one of them. That said, the book's source code contains as many aggregations as possible, especially those not discussed in this chapter.

> **NOTE** The code for this chapter is available on GitHub (http://mng.bz/wvZg) and on the book's website (https://www.manning.com/books/elasticsearch-in-action-second-edition).

13.1 Overview

Aggregations let companies understand their amassed data. They help understand customers and their relationships, evaluate product performance, forecast sales, and answer a wide range of questions about application performance over time, security threats, and more. Aggregations in Elasticsearch primarily fall into three categories:

- *Metric aggregations*—These aggregations generate metrics such as sum, average, minimum, maximum, top hits, mode, and many others. They include popular, single-value metrics that help understand data over a large set of documents by answering questions like: What is the sum of all product sales in the last month? What is the minimum number of API errors? and What are the top search hits?
- *Bucket aggregations*—Bucketing is a process of collecting data into interval buckets. These aggregations split the data into given sets: for example, splitting cars into years of registration, students into various grades, and so on. Histograms, ranges, terms, filters, and others fall into this category.
- *Pipeline aggregations*—These aggregations work on the output of other aggregations to provide a complex set of statistical analytics such as moving averages, derivatives, and others.

We discuss these aggregations with examples in the following sections. First, let's look at the syntax and endpoint used to carry out the aggregations.

13.1.1 The endpoint and syntax

After working with search queries, you should be pretty familiar with the `_search` endpoint. The good news is that we can use the same `_search` endpoint to run aggregations. However, the body of the request uses a new object: `aggregations` (`aggs` for short) instead of the usual `query` object. The following snippet shows the aggregation query syntax: